

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I Vaishnavi.s (192521313) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

Mock Interview

1. HDFS vs Traditional File Systems

HDFS (Hadoop Distributed File System) is a distributed file system designed to store and process very large datasets across multiple computers. Unlike traditional file systems that generally depend on a single server or storage device, HDFS distributes data across multiple DataNodes. It provides scalability, fault tolerance, and high-throughput data access. HDFS also divides large files into blocks and stores multiple copies of these blocks to protect against hardware failures.

2. Importance of Replication

Replication is the process of maintaining multiple copies of data on different storage nodes. It is important in distributed storage because hardware and network failures are common. If one node becomes unavailable, another node containing a replica can provide the required data. Replication therefore improves data availability, reliability, fault tolerance, and durability.

3. MapReduce

MapReduce is a distributed processing model used to process large datasets in parallel. The Map phase divides the input data into smaller portions and processes them to produce intermediate key-value pairs. The Shuffle and Sort phase groups the intermediate data according to keys. The Reduce phase processes these grouped values and generates the final output. This division allows large workloads to be processed efficiently across multiple machines.

4. Role of YARN

YARN (Yet Another Resource Negotiator) is the resource management layer of Hadoop. It manages the computing resources of a Hadoop cluster and allocates CPU and memory to different applications. It also schedules jobs, monitors their execution, and manages workloads across cluster nodes. YARN allows multiple processing frameworks and applications to share the same Hadoop infrastructure efficiently.

5. NAS and SAN

NAS (Network Attached Storage) provides file-level storage over a network and allows multiple users or systems to access shared files. It is commonly used for file sharing, backups, and document storage. SAN (Storage Area Network) provides block-level storage and presents storage resources to servers as disk volumes. SAN is generally used for high-performance enterprise applications such as databases. The main difference is that NAS works at the file level, while SAN works at the block level.

6. Purpose of ETL

ETL stands for Extract, Transform, and Load. It is used to collect data from different sources, clean and transform the data into a consistent format, and load it into a target storage or analytics system. ETL improves data quality and makes information suitable for reporting, analysis, and decision-making. In big data environments, ETL can handle data from databases, applications, logs, sensors, and other sources.

7. Apache NiFi

Apache NiFi is a data integration and flow management platform used to automate the movement of data between different systems. It provides a visual interface for designing data flows and supports data collection, routing, transformation, filtering, and delivery. NiFi also provides monitoring and data provenance, which helps organizations track the movement and history of data. It is useful for both real-time and batch data integration.

8. Failure Points in Hadoop Clusters

A Hadoop cluster can experience failures in DataNodes, disks, NameNodes, networks, and computing resources. DataNode failures can cause data to become temporarily unavailable, while disk failures can result in corrupted or lost blocks. NameNode failure can affect access to the file system metadata. Hadoop handles these problems through data replication, NameNode high availability, automatic recovery, monitoring, and re-replication of lost data blocks. These mechanisms help maintain cluster availability and reliability.

9. Secure Data Ingestion Pipeline

A secure data ingestion pipeline collects data from various sources and transfers it to a centralized storage or processing environment while protecting it from unauthorized access. Security should include authentication, authorization, encryption, data validation, access control, network security, and monitoring. Encryption protects data during transmission and storage, while role-based access control ensures that only authorized users can access sensitive information. Logging and auditing help identify suspicious activities and maintain accountability.

10. Scalable Hadoop-Based Processing System

A scalable Hadoop system processes large amounts of data by distributing both storage and computation across multiple machines. HDFS stores data across several nodes, while YARN manages the available computing resources. Processing frameworks such as MapReduce divide large jobs into smaller tasks that can execute simultaneously on different machines. When the amount of data increases, additional machines can be added to the cluster. This makes Hadoop suitable for organizations that need to process growing volumes of data efficiently and reliably.

